

Lab_02

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QUESTÃO 1

Para calcular o percentual de voos que tiveram atraso na chegada superior a 10 minutos, precisamos de número total de voos e número de voos atrasados ($\text{ARRIVAL_DELAY} > 10$).

QUESTÃO 2

```
getStats <- function(input, pos) {  
  input %>%  
    filter(AIRLINE == "AA" |  
           AIRLINE == "DL" |  
           AIRLINE == "UA" |  
           AIRLINE == "US") %>%  
    filter(!is.na(ARRIVAL_DELAY)) %>%  
    group_by(MONTH, DAY, AIRLINE) %>%  
    summarise(  
      total = n(),  
      atrasados = sum(ARRIVAL_DELAY > 10)  
    )  
}
```

QUESTÃO 3

```
resultado <- read_csv_chunked(  
  "flights.csv.zip",  
  callback = DataFrameCallback$new(getStats),  
  chunk_size = 1e6,  
  col_types = cols_only(  
    # ...  
  )  
)
```

```

    MONTH = col_integer(),
    DAY = col_integer(),
    AIRLINE = col_character(),
    ARRIVAL_DELAY = col_double()
  )
)

```

``summarise()`` has regrouped the output.
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i Summaries were computed grouped by MONTH, DAY, and AIRLINE.
i Output is grouped by MONTH and DAY.
i Use ``summarise(.groups = "drop_last")`` to silence this message.
i Use ``summarise(.by = c(MONTH, DAY, AIRLINE))`` for per-operation grouping
 (``?dplyr::dplyr_by``) instead.

QUESTÃO 4

```

computeStats <- function(stats) {

  stats %>%
    group_by(MONTH, DAY, AIRLINE) %>%
    summarise(
      total = sum(total),
      atrasados = sum(atrasados)
    ) %>%
    mutate(
      Perc = atrasados / total,
      Data = as.Date(paste(2015, MONTH, DAY, sep = "-"))
    ) %>%
    select(Cia = AIRLINE, Data, Perc)
}

stats_final <- computeStats(resultado)

```

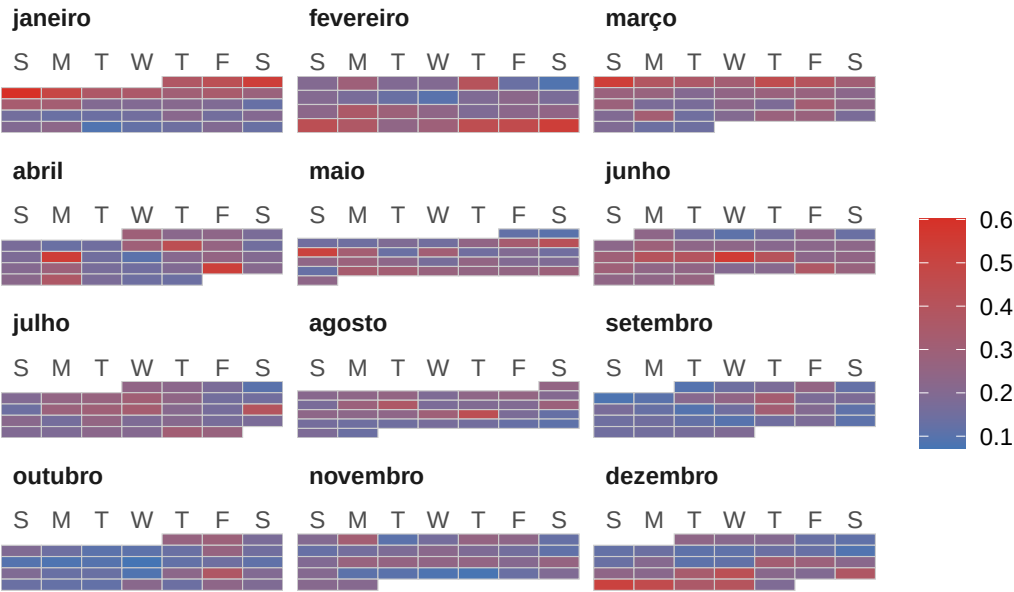
``summarise()`` has regrouped the output.

Adding missing grouping variables: `MONTH`, `DAY`
i Summaries were computed grouped by MONTH, DAY, and AIRLINE.
i Output is grouped by MONTH and DAY.
i Use `summarise(.groups = "drop\_last")` to silence this message.
i Use `summarise(.by = c(MONTH, DAY, AIRLINE))` for per-operation grouping
(`?dplyr::dplyr\_by`) instead.

QUESTÃO 5

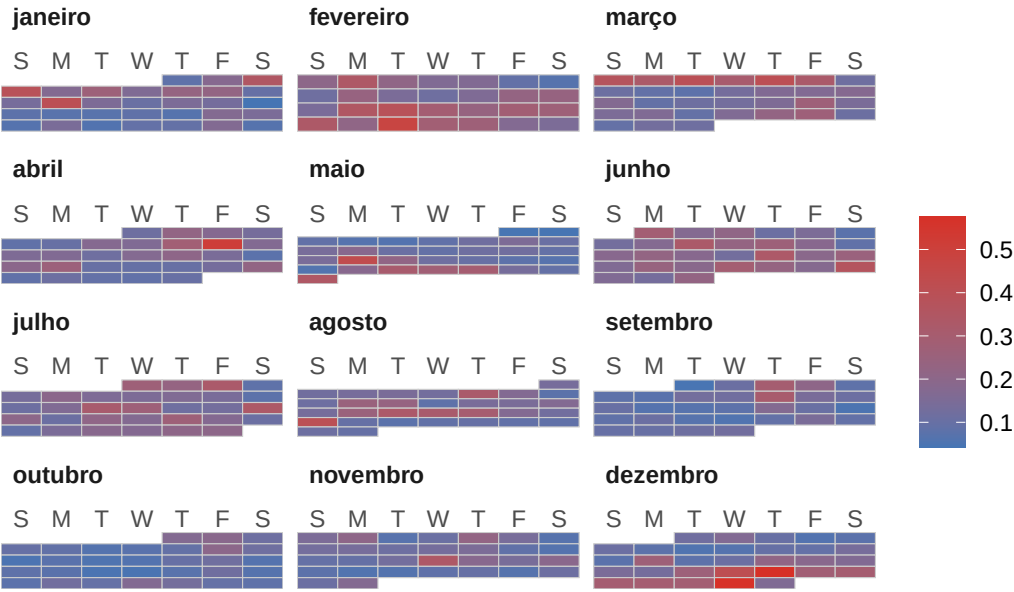
```
paleta <- scale_fill_gradient(  
  low = "#4575b4",  
  high = "#d73027"  
)  
  
baseCalendario <- function(stats, cia) {  
  
  dados <- stats %>%  
    filter(Cia == cia)  
  
  ggcal(dados$Data, dados$Perc)  
}  
  
cAA <- baseCalendario(stats_final, "AA")  
cDL <- baseCalendario(stats_final, "DL")  
cUA <- baseCalendario(stats_final, "UA")  
cUS <- baseCalendario(stats_final, "US")  
  
cAA + paleta + ggtitle("Atrasos - AA")
```

Atrasos – AA



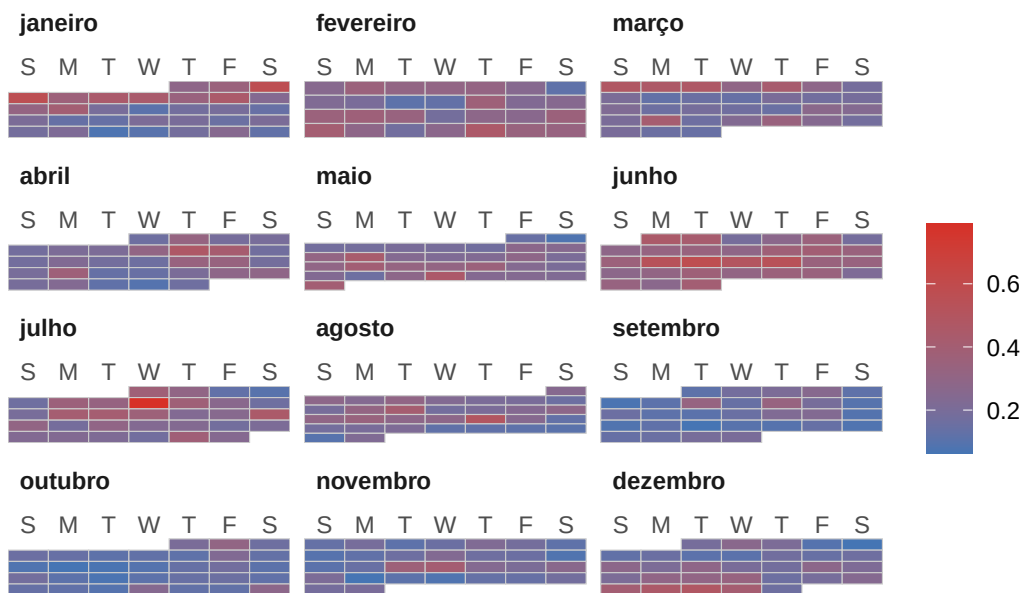
```
cDL + paleta + ggtitle("Atrasos - DL")
```

Atrasos – DL



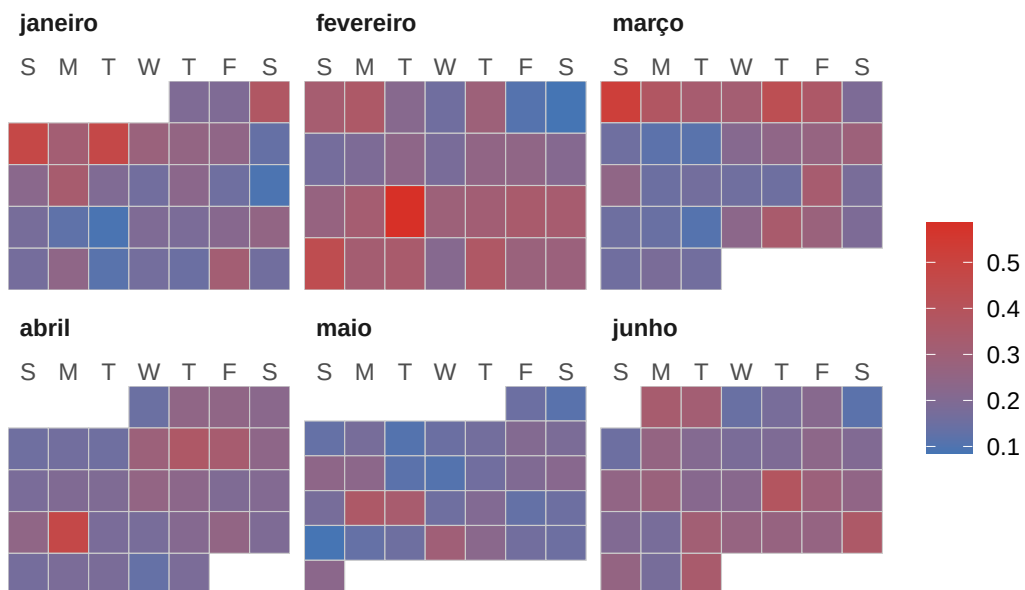
```
cUA + paleta + ggtitle("Atrasos - UA")
```

Atrasos – UA



```
cUS + paleta + ggtitle("Atrasos - US")
```

Atrasos – US



```
Sys.setenv(TZ = "America/Sao_Paulo")  
Sys.time()
```

```
[1] "2026-08-24 23:27:50 -03"
```